

AUTOMATE YOUR HOME WITH AI/ML-BASED SMART SPEAKERS

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Gone are the days when a speaker was just used for playing music. Today, you can command such smart home devices as smart plugs, smart bulbs/switches, smart locks and door cameras without even getting up, using a smart speaker. From streaming your favourite music and entertainment content on the speaker or compatible TV to activities like getting news updates, setting daily reminders and voice messaging, most speakers enable multiple actions with routine and hands-free control. With voice control, just say wake up words such as "Alexa" or "OK, Google" depending on your smart assistant, and it will instantly follow your requests.

The advantage of artificial intelligence (AI) and machine learning (ML) is that the speaker gets additional capabilities with time as the assistant learns new skills. Because it learns through experience, the more it is used, the smarter it becomes. You can still build a home theatre setup using large speakers and soundbars, or connect to a home media server.

Important features

There are many important parameters to consider before picking up a smart speaker that suits your requirements. Some

of these features are discussed next.

Sound quality. To ensure superior performance, basic functionality of the speaker—sound quality—needs to be a priority. Audio must be stable and clear in multiple rooms for not only proper music quality but to also control other devices amidst high background sounds.

Voice assistant. A voice assistant is the brain behind smart speakers. While some high-end speakers support multiple assistants, you need to choose your speaker based on which assistant works the best for you. Popular ones are Amazon Alexa, Apple Siri, Google Assistant and Microsoft Cortana. The ability to distinguish the voices of different users enables personalised content for each individual.

Far-field voice-recognition feature improves listening capability. Some smart speakers even support such local languages as Hindi.

Wireless connectivity. Smart speakers use either Bluetooth or Wi-Fi to connect to your smart devices. You need to consider parameters such as range before your device drops the signal.

When it comes to Bluetooth, most devices use Bluetooth 4.1. For Wi-Fi, dual-band (2.5GHz/5GHz) ensures faster speed.

Control smart home devices. The best part about a smart speaker is its ability to automate your home. It is therefore necessary to ensure that the one you are buying is compatible with the smart devices of various brands installed in your home. You also need to sync and connect your items. Only then you can access multiple features at once and broadcast messages to other smart devices. If the smart speaker is Chromecast-compatible, you will be able to access content from such services as

Netflix and Google Play Store through your smartphone easily.

Also, check the streaming services that are supported.

Power. If you want your speaker to be portable without losing connection, checking not just the size but also whether it uses a charger to power the battery is important.

Build. A speaker's build should be such that it is safe and reliable to use. Smart speakers are usually small and lightweight for portability. Some are even waterproof and feature a touchscreen. These have a fabric outer shell to provide high-quality sound as well as hear your voice (instructions) easily. Since these are fragile, it is essential to place them where these are safe and easy to access.

Customisation options are also available on some models.

Market options

There are several smart speakers available in the market from big brands, with Google and Amazon having the biggest market share. With the market growing rapidly, such options as Bixby-powered Samsung Galaxy Home Speaker are also coming up shortly. Some popular smart speakers are listed in this section. Prices may vary on different e-commerce websites.

Google Home Mini. Google Home Mini is a voice-controlled, hands-free smart speaker enabled by Google Assistant, which supports Hindi language, too. It measures 9.78cm x 4.2cm

